

1 Simulating the water

Variables and their meanings:

1. H : average water level (where the tank is filled to)
2. h : wave height (the real water height is $h+H$)
3. u : horizontal water velocity in the x-direction
4. v : horizontal water velocity in the y-direction
5. P_{ext} : external pressure applied to the water surface

Shallow Water Equations:

$$\frac{\partial h}{\partial t} + \frac{\partial(H+h)u}{\partial x} + \frac{\partial(H+h)v}{\partial y} = 0 \quad (1)$$

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + g \frac{\partial h}{\partial x} + ku + \frac{1}{\rho} \frac{\partial P_{ext}}{\partial x} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (2)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + g \frac{\partial h}{\partial y} + kv + \frac{1}{\rho} \frac{\partial P_{ext}}{\partial y} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) \quad (3)$$

Note: P_{ext} is the only term which drives the water. Additionally, ν is very small, $\rho \approx 1000$, and $k \approx 1$ for water.

1.1 Updating the fields through time

To simulate the differential equations, the differentials can be approximated to tiny intervals.

For the terms of the form $\frac{\partial f}{\partial t}$ for some f , they were used to update the field f from one frame to the next. i.e. using $\{f(t), f(t - \Delta t), f(t - 2\Delta t), \dots\}$ to determine $f(t + \Delta t)$.

$$\frac{\partial f}{\partial t} \approx \frac{\Delta f}{\Delta t} = \frac{f(t + \Delta t) - f(t)}{\Delta t} \quad (4)$$

But only using one piece of (potentially noisy) information, $f(t)$, makes the simulation really susceptible to exploding.

To make it more stable, $f(t - \Delta t)$ will also be used.

$$\frac{\partial f}{\partial t} \approx \frac{3s_1 + 1s_2}{4} \quad (5)$$

With s_1 the slope of f from the current frame to the last, and s_2 the slope from the last frame to the one before it. Therefore

$$\frac{\partial f}{\partial t} \approx \frac{3 \cdot \frac{f(t+\Delta t) - f(t)}{\Delta t} + 1 \cdot \frac{f(t) - f(t-\Delta t)}{\Delta t}}{4} = \frac{h(t + \Delta t) - \frac{2}{3}h(t) - \frac{1}{3}h(t - \Delta t)}{\frac{4}{3}\Delta t} \quad (6)$$

Given an equation of the form

$$\frac{\partial f}{\partial t} = A \quad (7)$$

$$\frac{h(t + \Delta t) - \frac{2}{3}h(t) - \frac{1}{3}h(t - \Delta t)}{\frac{4}{3}\Delta t} \approx A \quad (8)$$

The recursive formula to update f is

$$h(t + \Delta t) = \frac{4}{3}\Delta t \cdot A + \frac{2}{3}h(t) + \frac{1}{3}h(t - \Delta t) \quad (9)$$

1.2 Approximating spacial derivatives

The spacial derivatives can be approximated in a similar way, this time taking two slopes on each side.

$$\frac{\partial f}{\partial x} = \frac{s_1 + 3s_2 + 3s_3 + s_4}{8} \quad (10)$$

$$\frac{\partial f}{\partial x} = \frac{-f(x - 2\Delta x) - 2f(x - \Delta x) + 0f(x) + 2f(x + \Delta x) + f(x + 2\Delta x)}{8\Delta x} \quad (11)$$

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1.3 Implementing boundaries in the simulation

The simulation features reflective boundaries. To do this, the horizontal velocity near the borders was forced to gradually become 0 with distance to the border. Additionally, if values would have been sampled outside of the simulation's borders, the point gets reflected across the border and back into the simulated area by however much it would have "overshot".

2 Simulating the lasing medium

Since the water surface is flat with vertical waves, we'll consider the electric field as two-dimensional, and assume its vector at each point in space is pointing in the up-down direction.

We imagine an electron as a particle around an atom. We imagine the attraction to the nucleus as a restoring force in the form $-kz$ (like a spring force). Being a charged particle with charge $-e$, it is affected by the electric field $\mathbf{E}(\mathbf{r}, t)$ at its position $\mathbf{r} = (r_x, r_y)$ and at time t .

Since the electric field will always be in the up-down ($\hat{z}$) direction, the electron's oscillation will only happen in that direction.

Writing Newton's second law for the electron:

$$\mathbf{F}_{\text{net}} = m\ddot{\mathbf{z}} \quad (12)$$

$$-kz + q\mathbf{E}(\mathbf{r}, t) = m\ddot{\mathbf{z}} \quad (13)$$

$$(14)$$

$$\frac{q}{m}\mathbf{E} = \ddot{\mathbf{z}} + \frac{k}{m}z \quad (15)$$

This is the form of a driven harmonic oscillator. It is not so simple, though, because the electron, being a charged particle, affects the electric field as it is moved. For example, a wave pulse comes by in the up ($+z$) direction. This leads the electron to get pushed down (since its charge is negative). The electron is now below the 2d plane of the electric field. Then, since it's a charged particle, it pulls the field toward it, downward.

In a way, the electron can be seen to be damping its own motion through its interaction with the electric field. It edits the field against its position.

¹ $f(x)$ will always have a coefficient of 0 whenever the approximation is taken with equal weighting on the left and right of the point. This was deemed acceptable, as the approximation being symmetric was assumed to be more important.